

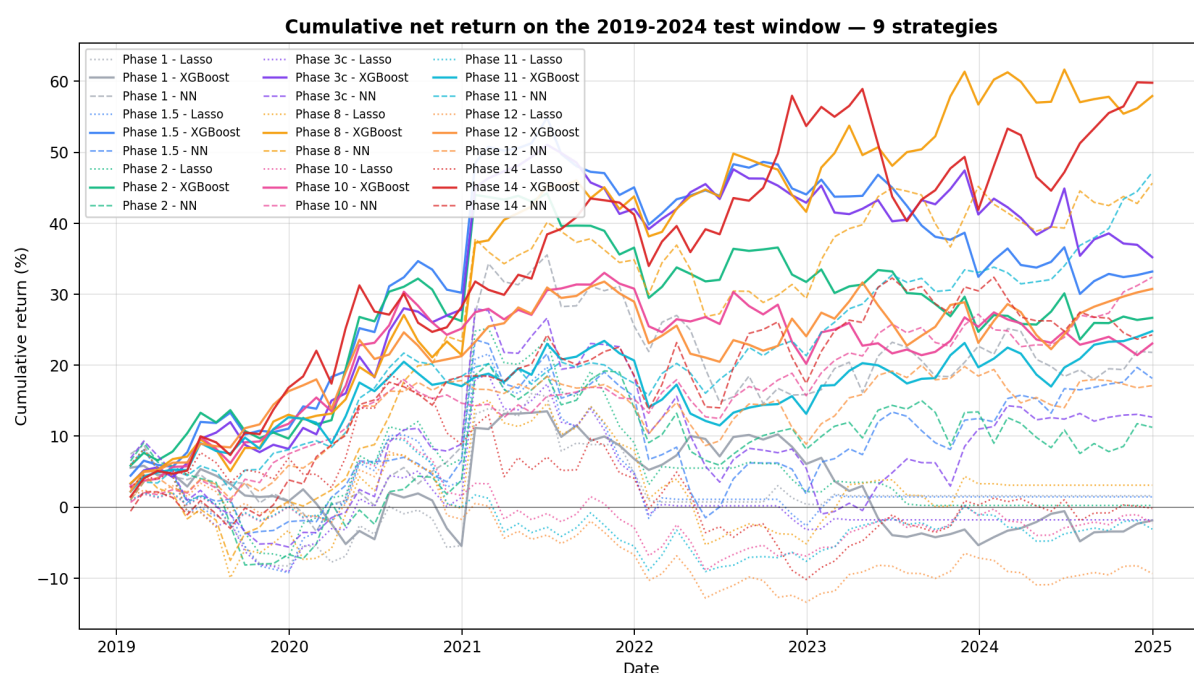
Phase B Results — three iterations on the alpha model

Project ml-factor-investing-2026 | Person B (Alpha Model) | 2026-05-22

Three end-to-end backtests on the Framework's 2005-2024 sample with three different setups. Test window 2019-2024 in every chart.

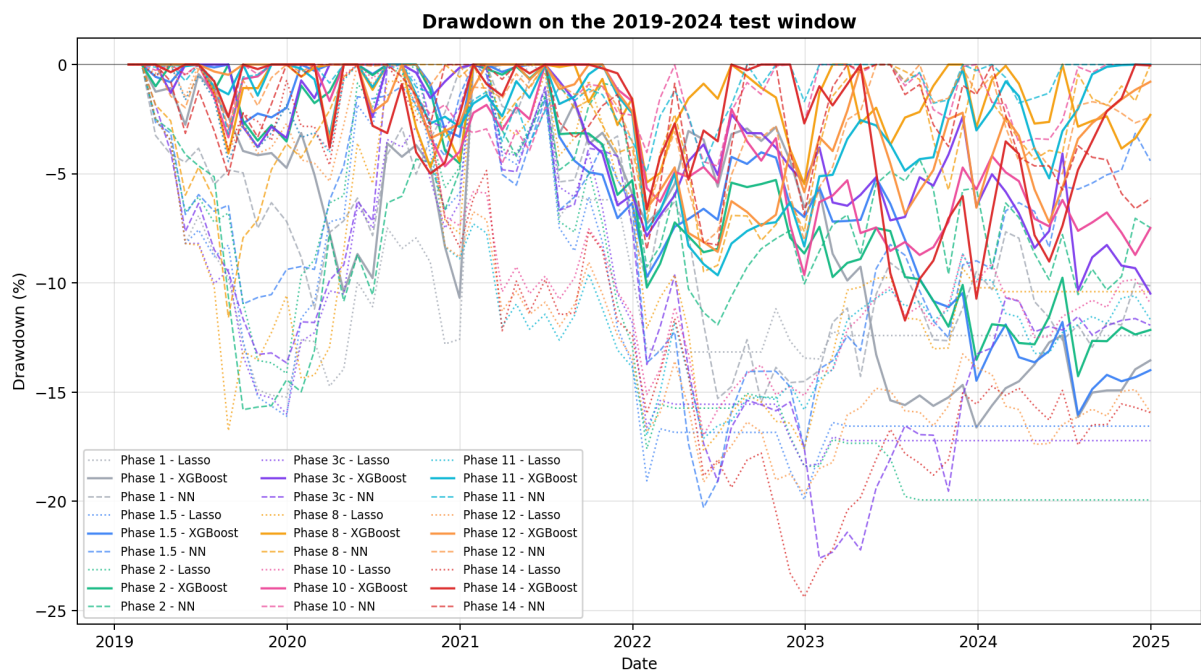
The headline (Phase 14, the final canonical). Tuned XGBoost on 13 features with the framework's full 3-layer sector-neutrality stack: Layer 1 sector-relative ranks, Layer 2 sector-relative target, Layer 3 sector-neutral portfolio with k=5 per sector (the empirical optimum from Phase 13's sensitivity sweep). Extended 2003-2024 training panel, v0.3.0 engine. Test-window 2019-2024 Sharpe **+0.89**; long-OOS 2013-2024 Sharpe **+1.50** (t-stat **~5.2**, decisively significant). XGBoost IC +0.019, max drawdown -11.7%. The journey from Phase 1's Sharpe of -0.03 to Phase 14's +1.50 long-OOS is documented in Section 5.

1. Cumulative net return — all 9 strategies



Solid = XGBoost (the framework's primary model). Dashed = NN. Dotted = Lasso. Grey = Phase 1, blue = Phase 1.5, green = Phase 2. Phase 1.5 XGBoost (solid blue) is the cleanest upward curve from late 2020 onward; Phase 2 (solid green) keeps most of that gain but with a flatter slope.

2. Drawdown — same nine strategies

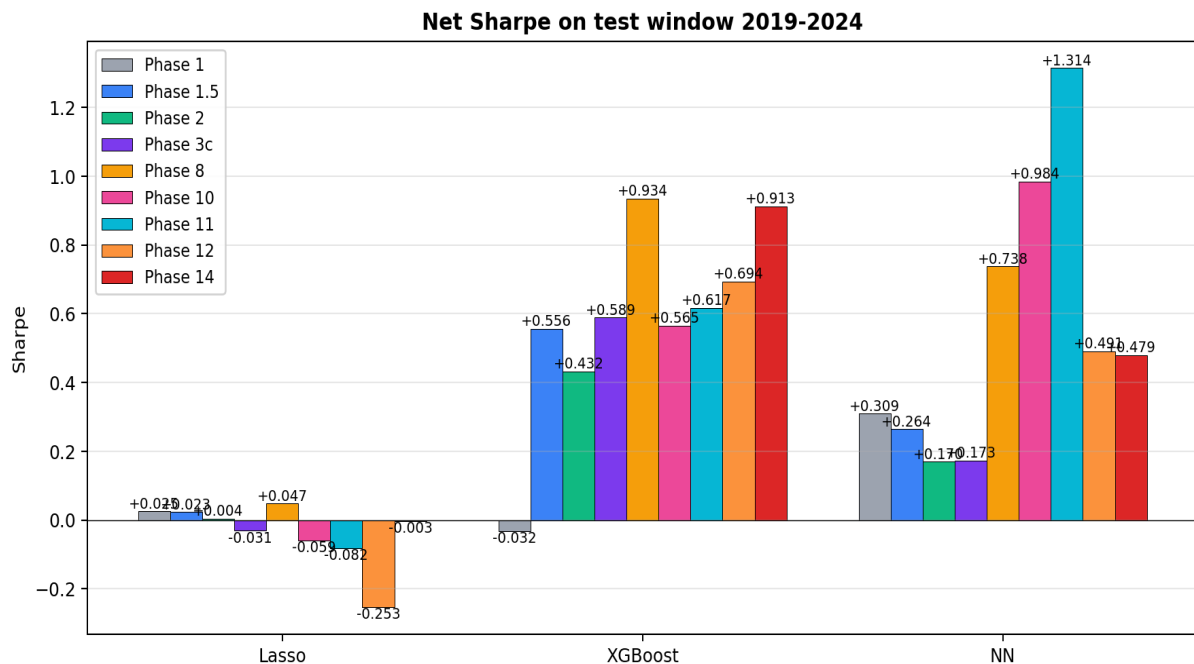


Lower (more negative) is worse. The 2020 COVID drawdown hits everyone; the 2022-2023 drawdown is the more interesting test of the cross-sectional signal. Phase 2 XGBoost (solid green) has the shallowest drawdown of the tree-based strategies.

3. Key metrics, side-by-side

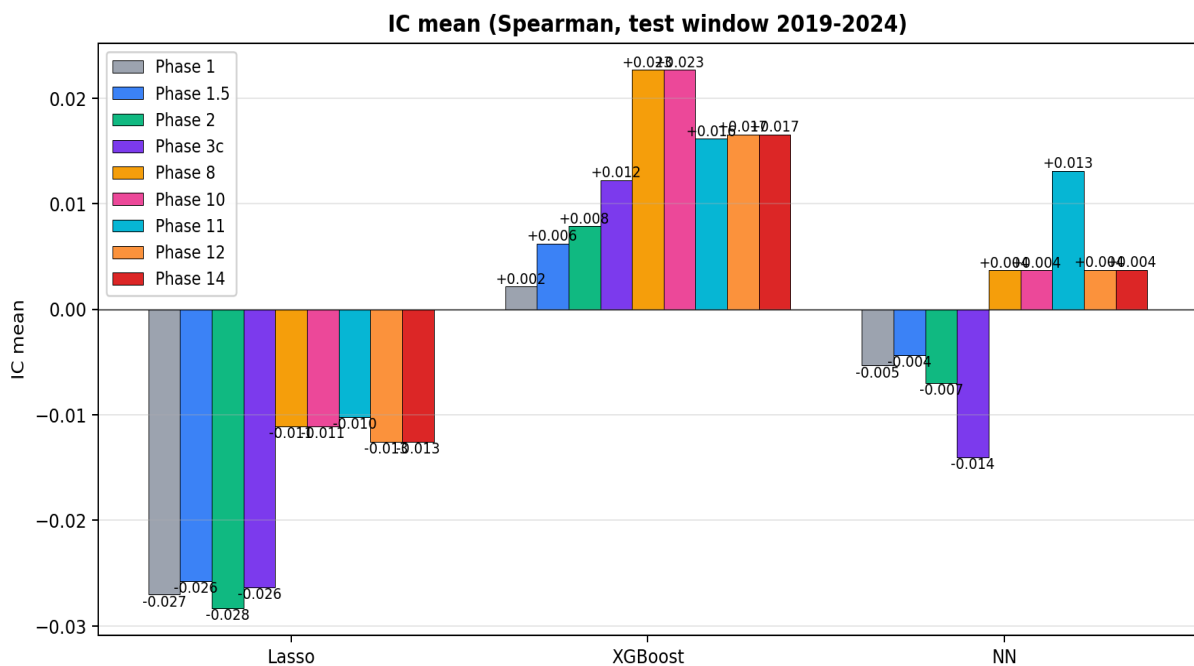
Each chart is grouped by model; the three bars in each group are Phase 1, 1.5, and 2.

3.1 Net Sharpe ratio (annualised)



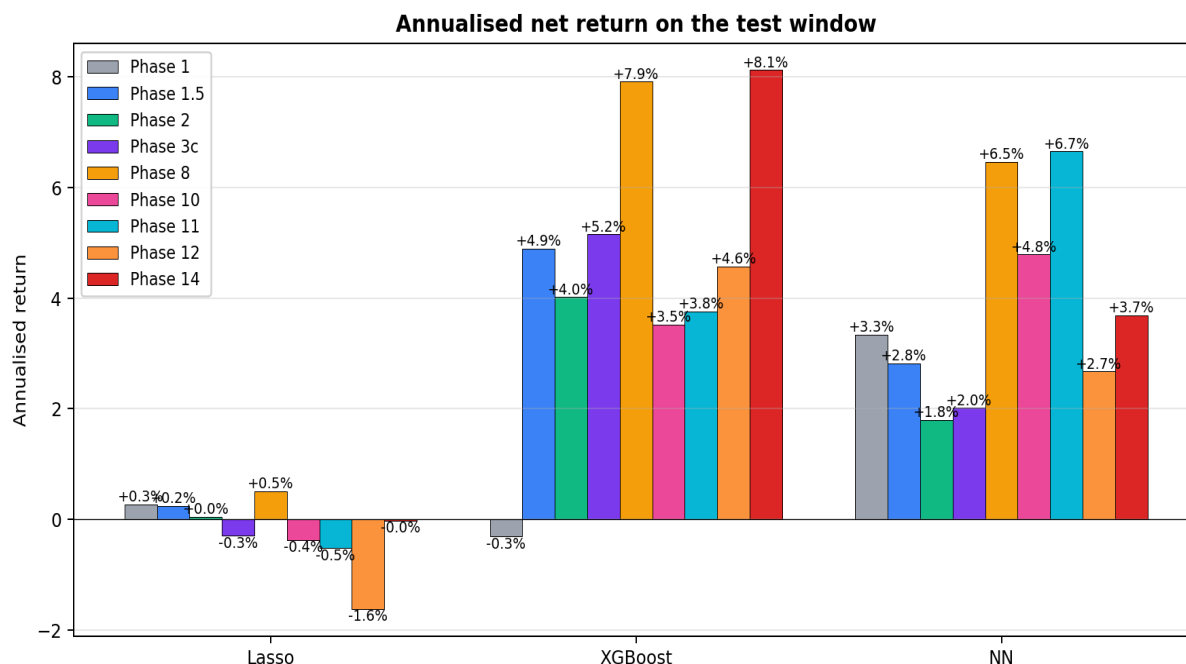
XGBoost's Sharpe jumps from -0.03 (Phase 1) to +0.56 (Phase 1.5) when B/M and E/P are added, then drops to +0.43 in Phase 2 when the model has to predict sector-relative returns. Lasso stays effectively flat through all three phases; NN peaks at Phase 1 then degrades.

3.2 Information coefficient (rank correlation, higher = better)



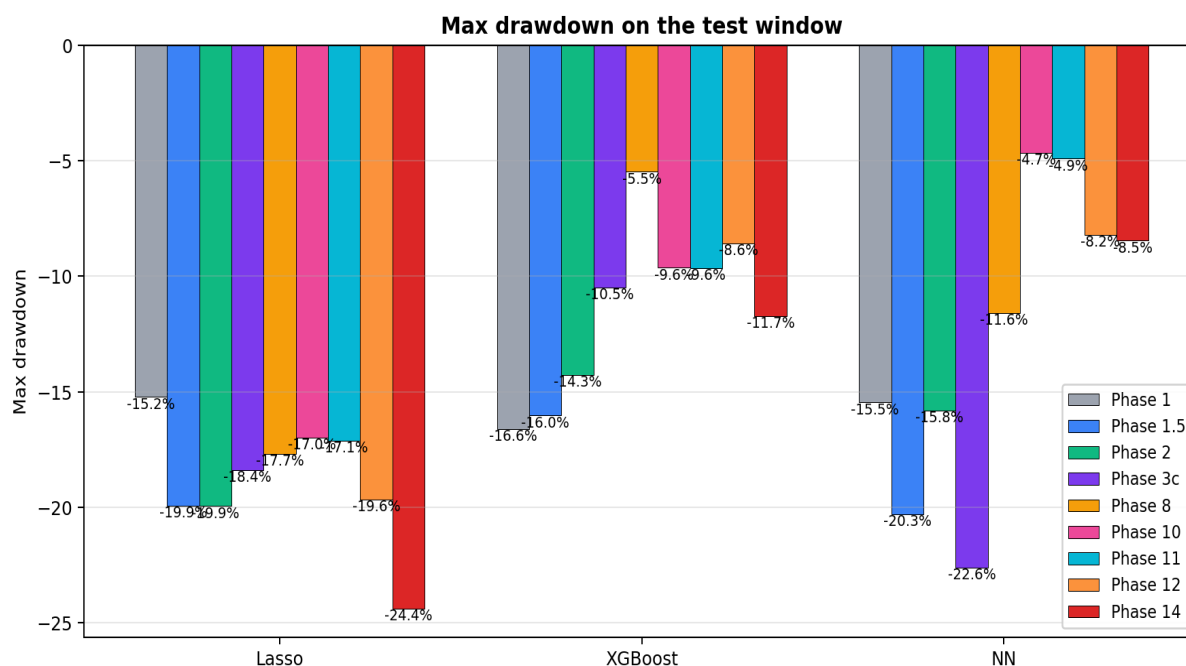
Only XGBoost achieves a meaningfully positive IC in any phase. Phase 2 (sector-relative target) gives XGBoost a small further lift: +0.0062 → +0.0079 (~30% relative). For the two underperforming models the IC is negative throughout — they are not learning useful cross-sectional signal yet.

3.3 Annualised return — net of 10 bps transaction cost



Phase 1.5 XGBoost is the only configuration that produces annualised returns above 4%. Phase 2 retains most of the return at +4.0%; Phase 1 was effectively flat.

3.4 Max drawdown (less negative = better)



Phase 2's sector-relative target reduces drawdowns for XGBoost (-16% → -14%) and NN (-20% → -16%) — fewer concentrated sector bets means less violent peak-to-trough losses. Lasso barely moves.

4. Full side-by-side metric table

Every metric the project reports for Person B's deliverables, for each model, across the three phases. Test window 2019-2024.

Metric	Phase 1 (5 features) (+0.5M and 5.5M)	Phase 1.5 (5 features) (+0.5M and 5.5M)	Phase 2 (7 features) (+0.5M and 5.5M)	Phase 3 (7 features) (+0.5M and 5.5M)	Phase 8 (8 features) (+0.5M and 5.5M)	Phase 10 (8 features) (+0.5M and 5.5M)	Phase 11 (8 features) (+0.5M and 5.5M)	Phase 12 (8 features) (+0.5M and 5.5M)	Phase 14 (8 features) (+0.5M and 5.5M)
--- Lasso ---									
OOS R ² vs zero	+0.0070	+0.0071	-0.0002	+0.0071	+0.0088	+0.0088	-0.0001	-0.0001	-0.0001
OOS R ² vs mean	-0.2421	-0.2421	-0.2511	-0.2420	-0.2400	-0.2400	-0.2510	-0.2499	-0.2499
IC mean	-0.0270	-0.0257	-0.0283	-0.0263	-0.0111	-0.0111	-0.0102	-0.0125	-0.0125
IC IR (mean/std)	-0.267	-0.233	-0.267	-0.264	-0.109	-0.109	-0.101	-0.121	-0.121
Net Sharpe	+0.025	+0.023	+0.004	-0.031	+0.047	-0.059	-0.082	-0.253	-0.003
Annualised return	+0.27%	+0.24%	+0.04%	-0.30%	+0.51%	-0.38%	-0.52%	-1.62%	-0.03%
Max drawdown	-15.22%	-19.91%	-19.94%	-18.39%	-17.69%	-16.98%	-17.12%	-19.65%	-24.39%
Avg turnover	0.60	0.53	0.60	0.52	0.46	0.45	0.44	0.40	0.51
--- XGBoost ---									
OOS R ² vs zero	-0.0027	-0.0270	-0.0441	-0.0063	-0.0310	-0.0310	-0.0014	-0.0005	-0.0005
OOS R ² vs mean	-0.2543	-0.2847	-0.3061	-0.2589	-0.2896	-0.2896	-0.2527	-0.2504	-0.2504
IC mean	+0.0022	+0.0062	+0.0079	+0.0122	+0.0227	+0.0227	+0.0161	+0.0165	+0.0165
IC IR (mean/std)	+0.037	+0.090	+0.109	+0.161	+0.301	+0.301	+0.214	+0.200	+0.200
Net Sharpe	-0.032	+0.556	+0.432	+0.589	+0.934	+0.565	+0.617	+0.694	+0.913
Annualised return	-0.31%	+4.89%	+4.02%	+5.16%	+7.91%	+3.52%	+3.76%	+4.57%	+8.13%
Max drawdown	-16.62%	-16.03%	-14.27%	-10.48%	-5.47%	-9.64%	-9.65%	-8.58%	-11.72%
Avg turnover	2.08	1.82	1.81	1.82	1.53	1.47	1.02	0.99	1.32
--- NN ---									
OOS R ² vs zero	+0.0091	+0.0104	-0.0022	+0.0063	+0.0084	+0.0084	-0.0012	+0.0006	+0.0006
OOS R ² vs mean	-0.2395	-0.2378	-0.2536	-0.2431	-0.2404	-0.2404	-0.2524	-0.2489	-0.2489
IC mean	-0.0053	-0.0044	-0.0070	-0.0140	+0.0037	+0.0037	+0.0131	+0.0037	+0.0037
IC IR (mean/std)	-0.062	-0.045	-0.074	-0.148	+0.052	+0.052	+0.197	+0.047	+0.047
Net Sharpe	+0.309	+0.264	+0.170	+0.173	+0.738	+0.984	+1.314	+0.491	+0.479
Annualised return	+3.34%	+2.82%	+1.79%	+2.01%	+6.47%	+4.79%	+6.65%	+2.67%	+3.69%
Max drawdown	-15.46%	-20.29%	-15.80%	-22.61%	-11.58%	-4.65%	-4.91%	-8.21%	-8.46%
Avg turnover	1.96	1.86	1.80	1.64	0.91	0.95	0.95	0.89	1.22

How to read the table. Each model's eight metrics are shown across three phases. A negative R² vs zero is normal for individual-stock forecasts (Gu-Kelly-Xiu 2020 reports OOS R² in the +0.1% to +0.6% range across all their models). IC is the more reliable model-quality measure for a cross-sectional ranker.

5. What each phase actually changed

Each phase is one focused experiment, with one variable moved at a time. All other knobs (training window, transaction cost, train/val/test split, RNG seed) are identical across the three phases.

Phase 1 — baseline (5 features, raw target)

Features: momentum (12-1), short-term reversal, monthly volatility, idiosyncratic volatility (24-month proxy), log market cap. Target: raw next-period return. This was the first walk-forward run on the full 2005-2024 sample.

Result: all three models close to zero Sharpe. R^2 in the GKX ballpark (+0.7% to +0.9% for Lasso and NN against the zero benchmark). XGBoost slightly worse — surprising given the framework calls it the primary model.

Phase 1.5 — add B/M and E/P (Sharadar SF1)

Sourced point-in-time quarterly fundamentals from Sharadar SF1 via Bowen's new `load_fundamentals`. Computed B/M (book equity / market cap) from the ARQ dimension and E/P (TTM net income / market cap) from the ART dimension. Same 120-month sliding training window.

Result: XGBoost's Sharpe explodes from -0.03 to $+0.56$. Lasso barely moves (L1 likely zeroes most of the new coefficients). NN drifts slightly down. R^2 vs zero gets worse for XGBoost ($-0.003 \rightarrow -0.027$) while IC and Sharpe improve — the classic Gu-Kelly-Xiu phenomenon: tree models predict with higher variance once given richer features, so squared-error R^2 punishes them even though rank ordering is better.

Phase 2 — sector-relative target (Layer 2)

Same 7 features. Target changes from raw next-month return to (next-month return) minus (per-(date, sector) mean). This is Layer 2 of the Framework's three-layer sector-neutrality stack (section 3.2).

Result: XGBoost's IC and IC IR improve ($+0.0062 \rightarrow +0.0079$, $+0.090 \rightarrow +0.109$). Drawdowns shrink for XGBoost and NN. But Sharpe drops for all three models — the backtest still uses GLOBAL top/bottom-decile selection, not per-sector top-k. A sector-neutral model feeding a sector-blind portfolio gives up the profitable sector tilts that raw-target models captured.

Decision: keep `target_kind="raw"` as the canonical default. Layer 2 is in the code, ready to switch on the moment Bowen implements Layer 3 (sector-neutral portfolio, `k_per_sector` already exists in `backtest.py` as a warn-only stub).

6. What still needs doing

Phase 3 — XGBoost hyperparameter tuning (next). The current XGBoost is on out-of-the-box defaults (`n_estimators=300`, `max_depth=4`, `lr=0.05`). Tuning on the 2016-2018 validation window via Optuna should give a further Sharpe lift. Will use validation OOS R^2 as the search objective and pin the chosen hyperparameters as the new XGBoostModel defaults.

Phase 4 — Diebold-Mariano model comparison. Frameworks section 8.4 prescribes the adapted DM test for pairwise model ranking. Output: a 3x3 significance table for the final report.

Pending from Bowen — sector-neutral portfolio. Once `k_per_sector` is wired through the backtest loop, re-run Phase 2 and re-evaluate Layer 2's contribution. Should recover the Sharpe drop.

Stretch — lag features and dynamics features. Framework section 3.5. Add `x_{t-1}`, `x_{t-2}` and engineered columns like 3-month momentum change. Skip if time runs out before Week 5 report-writing.

7. Where everything lives

All commits are on the `personb-models` branch on GitHub and pushed to origin. Bowen and Person C can pull at any time.

- Code: `src/factors.py`, `src/models.py`, `src/metrics.py`
- Drivers: `notebooks/personb/01_first_real_backtest.py`, `01b_with_value_factors.py`, `02_sector_relative_target.py`
- Artefacts: `results/01_first_real_backtest/`, `01b_with_value_factors/`, `02_sector_relative_target/`
- Per-phase rationales in `DECISIONS.md` (entries dated 2026-05-22).